

Structures, Functions & Procedures (Algo + C)

Statements with solutions in pseudo-code and C language

Exercise 1 — 2D Point and Distance to the Origin

Statement. Define `typedef struct {double x,y;} Point;` and write `double dist0(Point p)`.

Exercise 2 — Translation of a Point (Procedure)

Statement. Procedure `translater(Point* p, dx, dy)` that modifies `p`.

Exercise 3 — Rectangle: Area and Perimeter

Statement. `typedef struct {double L,l;} Rect;` Functions `aire(r)`, `peri(r)`.

Exercise 4 — Complex Numbers: Addition and Modulus

Statement. `typedef struct {double re,im;} Cx;` `add(a,b)`, `module(a)`.

Exercise 5 — Fraction: Normalize and Add

Statement. `typedef struct {int num,den;} Frac;` `pgcd`, `normalize`, `addf`.

Exercise 6 — Student: Init and Display

Statement. `typedef struct {char nom[32]; double note;} Etu;` `init`, `affiche`.

Exercise 7 — Best Student (Pointer Return)

Statement. Return the address of the student with the highest grade `best(E,n)`.

Exercise 8 — Sorting Students by Grade (Procedure)

Statement. `sort_note(E,n)` (bubble sort) ascending.

Exercise 9 — Time hh:mm:ss: Normalize and Add Seconds

Statement. `typedef struct {int h,m,s;} Time; norm, addsec.`

Exercise 10 — Date: Leap Year and Day of the Year

Statement. `typedef struct {int j,m,a;} Date; isLeap, dayOfYear.`

Exercise 11 — RGB $\rightarrow$ Grayscale Levels

Statement. `typedef struct {unsigned char r,g,b;} RGB; toGray.`

Exercise 12 — Quadratic Polynomial: Evaluation

Statement. `typedef struct {double a,b,c;} Poly2; eval(p,x).`

Exercise 13 — Circle: Surface and Point Inclusion

Statement. `typedef struct {Point c; double r;} Cercle; aire, contient.`

Exercise 14 — 2×2 Matrix and Matrix–Vector Product

Statement. `typedef struct {double a11,a12,a21,a22;} Mat2; mul(Mat2,Point).`

Exercise 15 — Book: Search by Author

Statement. `typedef struct {char titre[64], auteur[48]; int an;} Livre; findByAuthor.`

Exercise 16 — 3D Box: Volume and Nesting

Statement. `typedef struct {double L,l,h;} Box; volume, can_nest(a,b)` (a inside b without rotation).

Exercise 17 — Statistics: Min, Max, Mean, Variance

Statement. `typedef struct {double min,max,moy,var;} Stats; compute(T,n).`

Exercise 18 — Linked List: Push Front and Count

Statement. `typedef struct Node {int v; struct Node* next;} Node; push_front, length.`

Exercise 19 — Person with Birth Date: Age

Statement. `typedef struct {char nom[32]; Date dn;} Person; function age(Person p, Date ref).`

Exercise 20 — Bank Account: Secure Deposit/Withdrawal

Statement. `typedef struct {int id; double solde;} Compte; depot, retrait` (returns success/failure).